

THE RULER-AND-COMPASS PLANE IS NOT 5-COLOURABLE

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ABSTRACT. OpenAI recently proved that every colouring of the Euclidean plane with five colours gives the same colour to two points at distance 1, and formalized the proof in Lean. The proof works with the plane of algebraic points and assumes that every algebraic number of modulus 1 has roots of every order. We observe that square roots are enough. Hence the plane of points constructible with ruler and compass is not 5-colourable. These are the points of the smallest subfield of $\mathbb{C}$ closed under square roots. By the de Bruijn–Erdős theorem, some finite unit-distance graph with constructible vertices has no proper 5-colouring. Two observations suffice: a group-theoretic lemma about elements of order 2, and a choice of exponents in one multiple ergodic average that needs no shift. We checked the modified proof in Lean 4. The kernel accepts it, and only the standard axioms are used. A short argument on paper then extends the result to the plane over every Euclidean field, Archimedean or not: the real constructible numbers embed in every Euclidean field.

1. THE RESULT

The Hadwiger–Nelson problem asks for the least number $\chi(\mathbb{R}^2)$ of colours needed to colour the plane so that points at distance 1 get different colours. Isbell’s hexagonal colouring gives $\chi(\mathbb{R}^2) \leq 7$, and de Grey [4] proved $\chi(\mathbb{R}^2) \geq 5$; see Soifer [12] for the history. OpenAI [9] proved $\chi(\mathbb{R}^2) \geq 6$, with no regularity assumption on the colour classes. Their proof is not constructive, and we know of no finite unit-distance graph without a proper 5-colouring.

Identify the plane with $\mathbb{C}$. For a subset $S \subset \mathbb{C}$, a proper colouring of S is a colouring in which points of S at distance 1 get different colours. Let E_c be the field of *constructible numbers*: the smallest subfield of $\mathbb{C}$ closed under square roots, whose elements are the points constructible with ruler and compass from 0 and 1 [2, Ch. 10].

Theorem 1. *E_c has no proper 5-colouring.*

So the unit-distance graph on E_c has chromatic number 6 or 7. Larger planes contain E_c , so the theorem also holds for them. Examples are the origami numbers (closed under square and cube roots [1]), the numbers expressible by radicals, the algebraic numbers and $\mathbb{C}$ itself. Section 6 extends the theorem to the plane F^2 over every Euclidean field F .

Corollary 2. *Some finite set of constructible points has no proper 5-colouring.*

Proof. A graph is 5-colourable if all its finite subgraphs are, by the de Bruijn–Erdős theorem [3]. □

The coordinates of such a set lie in a field obtained from $\mathbb{Q}$ by finitely many square roots. We know of no such set. By contrast, the planes over $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ and $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ have chromatic number 4 [10, 5].

OpenAI’s proof has two steps:

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These results were obtained with the assistance of an AI system (Claude, by Anthropic), which modified the formal proof, ran the checks and helped to draft this note. The proof is OpenAI’s [9]; this note shows that it works over a smaller field. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

- [9, Prop. 4.2], the forward half of their transfer theorem, turns a proper colouring of the plane into a weak measurable colouring. Its proof first restricts the colouring to the countable plane E of algebraic points, and uses nothing else about it.
- [9, Thm. 1.4] shows that the plane has no weak measurable 5-colouring. It does not involve E .

We run the first step with E_c in place of the algebraic numbers. Section 2 lists what that step uses of its field. Sections 3 and 4 show that E_c has what it needs. Section 5 describes the formal proof.

Proof of Theorem 1. Suppose E_c had a proper 5-colouring. By Sections 2–4, the proof of [9, Prop. 4.2] goes through with E_c in place of the algebraic numbers. It gives a weak measurable 5-colouring of the plane, and [9, Thm. 1.4] excludes one. $\square$

2. WHAT THE PROOF USES OF THE FIELD

OpenAI work with $E = \overline{\mathbb{Q}} \subset \mathbb{C}$, its real points $F = E \cap \mathbb{R}$, and the group $K = \{u \in E : |u| = 1\}$ [9, Def. 2.1]. Their proof uses three properties of E :

- (E1) E is a countable subfield of $\mathbb{C}$, algebraic over $\mathbb{Q}$, closed under complex conjugation, and contains i . Since $\mathbb{Q}(i) \subset E$, E is dense in $\mathbb{C}$, F in $\mathbb{R}$ and K in the circle. These density facts are used in [9, Lemma 2.5, Lemma 4.1, Prop. 4.2].
- (E2) F is closed under square roots of its nonnegative elements. This is used in four places:
 - the decomposition $x = k + \bar{k}$ with $k = x/2 + i\sqrt{1 - x^2/4}$ in [9, Lemma 2.7];
 - the number $L = \sqrt{AB}$ in [9, Lemma 3.3];
 - the points $t + i\sqrt{1 - t^2}$ in [9, §3.5];
 - the moduli $|z| \in F$ of the elements $z \in E$.
- (E3) K is divisible. It is used in three places:
 - [9, Lemma 2.4] uses that a finite image of K is trivial;
 - [9, Lemma 3.2] uses that the power maps $u \mapsto u^n$ in its multiple averages are onto;
 - [9, Lemma 3.3] uses directly the square map, in (3.15), and $u \mapsto u^{-2}$, in (3.13).

For E_c , (E1) and (E2) hold, but (E3) holds only for the exponents $\pm 2^k$.

The map $u \mapsto u^2$ is onto K : the square roots of $a + bi$, with $a^2 + b^2 = 1$, have coordinates $\pm\sqrt{(1+a)/2}$ and $\pm\sqrt{(1-a)/2}$. The map $u \mapsto u^3$ is not onto. Take $u = (3+4i)/5 = (2+i)/(2-i)$. It has valuation 1 at the prime $2+i$ of $\mathbb{Z}[i]$, so it is not a cube in $\mathbb{Q}(i)$. Hence $z^3 - u$ is irreducible over $\mathbb{Q}(i)$, and its roots, of degree 3 over $\mathbb{Q}(i)$, are not constructible.

3. ELEMENTS OF ORDER TWO

In [9, Lemma 2.4], a coset $d + C$ of positive mass is fixed by some rotation $u \neq 1$. Then $(u-1)d \in C$, and so $d \in C$. The formalization takes $u = -1$. That -1 fixes the coset follows from the next lemma, applied to the action of K on the finite orbit of the coset.

Lemma 3. *Let G be a commutative group in which every element is a square, and let $f: G \rightarrow H$ be a homomorphism with finite image. Then $f(g) = 1$ for every $g \in G$ with $g^2 = 1$.*

Proof. Squaring maps the finite group $f(G)$ onto itself, because every element of G is a square. So squaring is injective on $f(G)$. Since $f(g)^2 = f(g^2) = 1 = 1^2$, it follows that $f(g) = 1$. $\square$

The same lemma also gives the second use of divisibility in the formal proof. In the null-set statement for lines in [9, Lemma 2.4], it shows that K is infinite: if K were finite, the identity map of K would send -1 to 1 .

4. THE TRIPLE AVERAGE AT THE EXPONENTS $-1, 0, 1$

We use the notation of [9, §3]:

- X is an ergodic standard probability-preserving K -system, and Y is a factor with $R = X \times_Y X$ ergodic, as in [9, Lemma 3.2];
- $P = \mathbb{E}(\cdot \mid Y)$;

- U_u is the Koopman operator of $u \in K$. It is unitary, multiplicative, and commutes with P ;
- averages are taken along a Følner sequence of K .

The radial inequality $m(A)m(B) \leq m(|A - B|)$ of [9, Lemma 3.3] rests on two facts:

- the identity $|A - Bu^2| = |-L + (A - B)u + Lu^2|$, for $L = \sqrt{AB}$ and $|u| = 1$;
- the multiple average of [9, Lemma 3.2] for the characters $\psi_{-L} \cdot U_u \psi_{A-B} \cdot U_{u^2} \psi_L$, at the exponents 0, 1, 2.

In the proof of [9, Lemma 3.2], at (3.9), a common integer is added to the exponents of each term so that none is zero. The formalization uses one shift for all terms and requires every shifted exponent to be onto. For 0, 1, 2, one of three consecutive nonzero integers is a nonzero multiple of 3, so this needs cube roots. A shift chosen separately for each term would avoid them; we avoid the shift altogether.

Since $|u| = 1$,

$$|-L + (A - B)u + Lu^2| = |-Lu^{-1} + (A - B) + Lu|,$$

and, since U_u is unitary, multiplicative and commutes with P ,

$$\|P(a \cdot U_u b \cdot U_{u^2} c)\| = \|P(U_{u^{-1}} a \cdot b \cdot U_u c)\|.$$

So it is enough to prove the conclusion (3.2) of [9, Lemma 3.2] at the exponents $-1, 0, 1$, where the middle factor is fixed.

Lemma 4. *Let X , Y and R be as above, and suppose that $u \mapsto u^2$ is onto K . For bounded functions a, b, c on X ,*

$$\text{Avg}_u \|P(U_{u^{-1}} a \cdot b \cdot U_u c) - U_{u^{-1}} P a \cdot P b \cdot U_u P c\|_2^2 \longrightarrow 0$$

along every Følner sequence.

Proof. Write each factor as $v = Pv + (v - Pv)$ and expand. The term with no centred factor is $U_{u^{-1}} P a \cdot P b \cdot U_u P c$. It is Y -measurable, so P fixes it.

Consider a term whose only centred factor is $b - Pb$. Its other two factors are Y -measurable, so its projection is $U_{u^{-1}} P a \cdot U_u P c \cdot P(b - Pb) = 0$.

In every remaining term a centred factor sits at the exponent -1 or 1 . Let $W_u = U_{u^{-1}} a' \cdot b' \cdot U_u c'$ be such a term. By conditional independence,

$$\|PW_u\|_2^2 = \int_R \Phi \cdot G_u d\mu_R, \quad \Phi = b' \otimes \bar{b}', \quad G_u = U_{u^{-1}}(a' \otimes \bar{a}') \cdot U_u(c' \otimes \bar{c}').$$

One of the two factors of G_u is centred over Y , because $\mathbb{E}(w \otimes \bar{w} \mid Y) = |Pw|^2$.

Statement (3.5) of [9], applied on R to the two distinct nonzero exponents -1 and 1 , says that $\text{Avg}_u G_u \rightarrow 0$ in $L^2(R)$. Its proof uses three facts:

- R is ergodic;
- the pair criterion (3.3) holds on R at $d = \pm 1$;
- the case $l = 1$ holds through the power maps $u \mapsto u^{\pm 2}$, which are onto.

The function Φ does not depend on u , so $|\text{Avg}_u \int_R \Phi G_u d\mu_R| \leq \|\Phi\|_2 \|\text{Avg}_u G_u\|_2 \rightarrow 0$. This is where it matters that the middle factor is fixed. There are finitely many terms, and the lemma follows. $\square$

The power maps used are $u \mapsto u^{\pm 1}$ and $u \mapsto u^{\pm 2}$. The other two averages in [9, Lemma 3.3] need only the square map:

- the pair at the exponents 0, 2 is the pair criterion (3.3) with $d = 2$;
- the pair at the exponents 1, -1 needs no shift, and its difference is 2.

With Lemmas 3 and 4, the rest of the proof of [9, Prop. 4.2] goes through over E_c unchanged. This completes the proof of Theorem 1.

5. THE FORMAL PROOF

OpenAI's formalization [9] uses Lean 4 [7] with Mathlib [8]; we used the same versions, Lean 4.34.1 and Mathlib at commit `d13f23b7`. The proof is the import closure of their file `Five.lean`: 70 modules of their library, on top of Mathlib. We changed it in two ways:

- a new file defines the field. It proves that the field is algebraic and closed under conjugation, contains i , and satisfies (E2). Countability and the square roots in K are derived from these facts in other files. Eleven files of the closure were adapted.
- Two new lemmas in `Energy.lean` prove Lemma 4, and the radial inequality uses it. Like the rest of the formal proof, they are stated for invariant means rather than Følner sequences, so Lean proves the invariant-mean form of Lemma 4.

The final statement writes the field out, so the reader does not need to trust our definitions. In abbreviated form:

```
theorem sqrtClosure_plane_not_five_colourable :
  ¬ ∃ c : (sInf {s : IntermediateField ℚ ℂ | ∀ x : ℂ, x² ∈ s → x ∈ s}) → Fin 5,
    ∀ x y, ‖(x : ℂ) - (y : ℂ)‖ = 1 → c x ≠ c y
```

`#print axioms` lists only `propext`, `Classical.choice` and `Quot.sound`. A second file proves Corollary 2 from it, with Mathlib's compactness theorem for graph homomorphisms; the axioms are the same.

We also checked the proof in two further ways:

- a fresh copy of OpenAI's sources (commit `fd4aeeb2`) was patched and built from scratch: the 72 modules of the closure, namely their 70, the field file and the statement file, with none failing. Mathlib was prebuilt;
- `leanchecker` replayed these 72 modules in the kernel. Mathlib was not replayed.

An AI-assisted review of an earlier version of these changes found no mathematical error.

Before the constructible case we built the origami and radical cases and checked their axioms. The origami case was also rebuilt from scratch with the published script. The patches, the statements and a build script are in `lean/external/openai-five/fields/` of [11], with a copy of OpenAI's licence (Apache 2.0).

6. EVERY EUCLIDEAN FIELD

A *Euclidean field* is an ordered field in which every positive element is a square. For a field F , call two points (x, y) and (x', y') of F^2 at distance 1 if $(x - x')^2 + (y - y')^2 = 1$.

Corollary 5. *Let F be a Euclidean field. Every colouring of F^2 with five colours gives the same colour to two points at distance 1.*

Proof. Let $F_c = E_c \cap \mathbb{R}$. It is the smallest subfield of $\mathbb{R}$ closed under square roots of positive elements. Indeed, it is closed under them: if $0 < a \in F_c$, then $\sqrt{a} \in E_c \cap \mathbb{R}$. Conversely, let $L \subseteq \mathbb{R}$ be a field closed under them. Then $L(i)$ is closed under square roots: the square roots of $a + bi$, with $a, b \in L$ and $r = \sqrt{a^2 + b^2}$, are $\pm(\sqrt{(r+a)/2} + i\varepsilon\sqrt{(r-a)/2})$, where $\varepsilon = 1$ if $b \geq 0$ and $\varepsilon = -1$ otherwise. So $E_c \subseteq L(i)$, and $F_c \subseteq L(i) \cap \mathbb{R} = L$. With $L = F_c$ this also gives $E_c = F_c(i)$.

Hence F_c is the union of the fields $\mathbb{Q}(\sqrt{a_1}, \dots, \sqrt{a_n}) \subseteq \mathbb{R}$ in which each a_j is a positive element of $\mathbb{Q}(\sqrt{a_1}, \dots, \sqrt{a_{j-1}})$: this union is a field, closed under square roots of positive elements, and contained in F_c . As F_c is countable, it is also the union of one chain $\mathbb{Q} = L_0 \subseteq L_1 \subseteq \dots$ with $L_j = L_{j-1}(\sqrt{a_j})$ and $0 < a_j \in L_{j-1}$. To build it, list the elements of F_c , and adjoin, one by one, the square roots that a field of the above form containing the next element uses.

We define order-preserving field homomorphisms $\sigma_j : L_j \rightarrow F$, each extending the previous one. Let σ_0 be the inclusion of $\mathbb{Q}$. Given σ_{j-1} , the element $\sigma_{j-1}(a_j)$ is positive, so it equals b^2 for some $b > 0$ in F . If $\sqrt{a_j} \in L_{j-1}$, let $\sigma_j = \sigma_{j-1}$. Otherwise each element of L_j is $u + v\sqrt{a_j}$ with unique $u, v \in L_{j-1}$, and we let $\sigma_j(u + v\sqrt{a_j}) = \sigma_{j-1}(u) + \sigma_{j-1}(v)b$. This is a field homomorphism, because $b^2 = \sigma_{j-1}(a_j)$. It preserves the order, by the following rule, valid in every ordered field for $a > 0$ and $s > 0$ with $s^2 = a$. The sign of $u + vs$ is the sign of u if $v = 0$ or if u and v have

the same sign; it is the sign of v if $u = 0$; and it is the sign of u times the sign of $u^2 - av^2$ if u and v are nonzero with opposite signs. Apply the rule to $s = \sqrt{a_j}$ in L_j and to $s = b$ in F . Since σ_{j-1} preserves the signs of u , v and $u^2 - a_j v^2$, and $\sigma_{j-1}(u)^2 - b^2 \sigma_{j-1}(v)^2 = \sigma_{j-1}(u^2 - a_j v^2)$, the elements $u + v\sqrt{a_j}$ and $\sigma_j(u + v\sqrt{a_j})$ have the same sign. Together the σ_j define a field homomorphism $\sigma: F_c \rightarrow F$.

The map $E_c = F_c(i) \rightarrow F^2$, $x + iy \mapsto (\sigma(x), \sigma(y))$, is injective. If $z = x + iy$ and $z' = x' + iy'$ in E_c satisfy $|z - z'| = 1$, then $(x - x')^2 + (y - y')^2 = 1$, and applying σ shows that their images are at distance 1. So a colouring of F^2 with five colours and no two points at distance 1 of the same colour would restrict to a proper 5-colouring of E_c , against Theorem 1. $\square$

Examples of Euclidean fields are $\mathbb{R}$, the real algebraic numbers, F_c itself and every real closed field, such as a field of real Puiseux series or a field of hyperreal numbers. OpenAI's proof is written for $\mathbb{R}^2$, with Lebesgue measure, but it shows that already the plane of algebraic points has no proper 5-colouring (Section 1). So if F contains a copy of the real algebraic numbers, as every real closed field does, the embedding argument applied to the algebraic plane gives the corollary without Theorem 1. If F contains no such copy, as for F_c , the corollary rests on Theorem 1; for F_c it is Theorem 1, since $E_c = F_c(i)$. Corollary 5 also holds with $(x - x')^2 + (y - y')^2 = c$ for any fixed $c > 0$ in F in place of 1: since F is Euclidean, $c = r^2$ with $r > 0$, and scaling by r maps the pairs at distance 1 onto these pairs. Finally, the Hilbert planes that satisfy the parallel axiom and the circle-circle intersection property, which Hartshorne calls Euclidean planes, are up to isomorphism the planes F^2 over Euclidean fields F [6]; congruence of segments becomes equality of $(x - x')^2 + (y - y')^2$. So in every such plane, whichever segment is taken as unit, the points cannot be 5-coloured without two points at unit distance of the same colour.

7. REMARKS

Property (E2) says that the field $F = E \cap \mathbb{R}$ of Section 2 is Euclidean, and the method needs it: with $B = 1$, [9, Lemma 3.3] uses $\sqrt{A}$ for every $A \geq 0$ in F . Every Euclidean subfield of $\mathbb{R}$ contains $F_c = E_c \cap \mathbb{R}$ (proof of Corollary 5). So E_c is the smallest field of the form $F(i)$ to which the proof applies in this form.

Two questions remain open:

- Does the plane over the Pythagorean closure of $\mathbb{Q}$ have a proper 5-colouring? There K is still 2-divisible, because $(1 \pm a)/2 = ((1 \pm a)/2)^2 + (b/2)^2$ is a sum of squares. Of (E1)–(E3), only (E2) fails there.
- How many square roots must be adjoined to $\mathbb{Q}$ to obtain a field whose plane has no proper 5-colouring? Corollary 2 says that finitely many suffice.

An explicit finite unit-distance graph with no proper 5-colouring would bound the answer to the second question. It would also give an explicit witness for $\chi(\mathbb{R}^2) \geq 6$.

Remark. Over the Pythagorean closure P of $\mathbb{Q}$, the smallest subfield of $\mathbb{R}$ that contains $\sqrt{a^2 + b^2}$ whenever it contains a and b , the failure of (E2) is not just a gap in the proof. We use the notation of [9, Def. 2.1 and 2.2] with $\overline{\mathbb{Q}}$ replaced by $E = P(i)$. Thus $F = E \cap \mathbb{R} = P$, $K = \{u \in E : u\bar{u} = 1\}$, D is the compact group of characters of the discrete group E , on which $a \in E$ acts by $(ad)(z) = d(az)$, and $C = j(\mathbb{R}^2)$, where $j(\eta)(z) = e^{i\eta \cdot z}$. A K -invariant Borel probability measure ν on D is wild if $\nu(C) = 0$, and its Fourier coefficients are $f_\nu(z) = \int_D d(z) d\nu(d)$. The analogue of [9, Thm. 2.3] for $E = P(i)$, that every wild K -invariant probability measure on D is Haar measure, is false. That theorem is applied only in the proof of [9, Prop. 4.2], to make such Fourier coefficients vanish, at the elements of K in particular. So the first step cannot be run over P as it stands. This does not decide the first question above.

The totally real algebraic numbers form a subfield of $\mathbb{R}$ closed under $(a, b) \mapsto \sqrt{a^2 + b^2}$: each conjugate of $\sqrt{a^2 + b^2}$ is $\pm\sqrt{\tau(a)^2 + \tau(b)^2}$ for some embedding τ of $\mathbb{Q}(a, b)$ into $\mathbb{R}$, hence real. So every element of P is totally real. As $\sqrt{2} = \sqrt{1^2 + 1^2} \in P$, the embedding of $\mathbb{Q}(\sqrt{2})$ into $\mathbb{R}$ with $\sqrt{2} \mapsto -\sqrt{2}$ extends to an embedding σ of the algebraic extension P into $\mathbb{C}$, and $\sigma(P) \subset \mathbb{R}$ because P is totally real. In particular P is not Euclidean: $\sqrt{2} - 1 > 0$, but $\sigma(\sqrt{2} - 1) < 0$, so

$\sqrt{2} - 1$ is not a square in P . Extend σ to E by $\sigma(i) = i$. Then $\sigma(\bar{z}) = \overline{\sigma(z)}$, so $|\sigma(u)| = 1$ for $u \in K$.

For $\xi \in \mathbb{R}^2$ let $j_\sigma(\xi) \in D$ be the character $z \mapsto e^{i\xi \cdot \sigma(z)}$. The map $j_\sigma: \mathbb{R}^2 \rightarrow D$ is continuous, and $u \in K$ maps $j_\sigma(\xi)$ to $j_\sigma(\xi')$, where ξ' is ξ rotated by $\sigma(u)^{-1}$. So for $\rho > 0$ the image ν under j_σ of the uniform probability measure on the circle $|\xi| = \rho$ is a K -invariant Borel probability measure on D . It is wild. Indeed, if $j_\sigma(\xi) = j_\sigma(\eta)$, the continuous character $(z, w) \mapsto e^{i(\xi \cdot w - \eta \cdot z)}$ of $\mathbb{C}^2$ is trivial on the set $\{(z, \sigma(z)) : z \in E\}$. This set is dense in $\mathbb{C}^2$, because $x \mapsto (x, \sigma(x))$ maps $\mathbb{Q}(\sqrt{2})$ onto a dense subset of $\mathbb{R}^2$, the image of $\mathbb{Q}^2$ under the invertible linear map $(a, b) \mapsto (a + b\sqrt{2}, a - b\sqrt{2})$. So $\xi = \eta = 0$. Hence $j_\sigma^{-1}(C) = \{0\}$, which the circle misses, and $\nu(C) = 0$. But ν is not Haar measure. Its Fourier coefficient at z is $J_0(\rho|\sigma(z)|)$, where $J_0(r) = \frac{1}{2\pi} \int_0^{2\pi} e^{ir \cos \theta} d\theta$. At every $u \in K$ it is $J_0(\rho)$, which is not zero for $0 < \rho < 2.4$, whereas the Fourier coefficients of Haar measure vanish at every $z \neq 0$.

By contrast, a Euclidean subfield of $\mathbb{R}$ has no embedding into $\mathbb{R}$ other than the inclusion: an embedding maps squares to squares, so it preserves the order, and it fixes $\mathbb{Q}$, which is dense in $\mathbb{R}$. So for $\overline{\mathbb{Q}}$ and for E_c an embedding of E that commutes with complex conjugation is the identity or complex conjugation, $j_\sigma(\mathbb{R}^2) = C$, and the construction gives nothing.

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